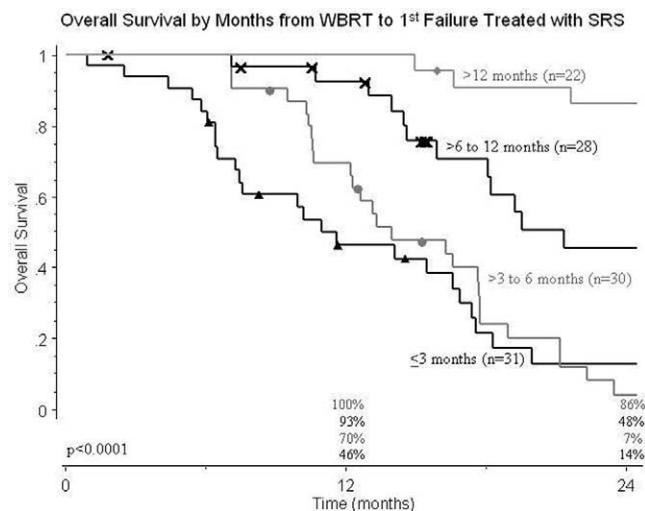




1021 Phase I Trial of Arsenic Trioxide and Fractionated Stereotactic Radiation for Recurrent Malignant Glioma

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Purpose/Objective: We have previously shown that arsenic trioxide (ATO) combined with radiation enhances the response of human malignant glioma in vitro and in vivo in a preclinical model. We are conducting a phase I trial of ATO and fractionated stereotactic radiation in patients with recurrent malignant glioma to determine the maximum tolerated dose and to define the toxicity profile in this population.

Materials/Methods: Patients were enrolled to this IRB-approved, open label phase I dose-escalation trial. Eligibility criteria included histologically confirmed malignant glioma in adults ≥ 18 years of age with Karnofsky Performance Status ≥ 60 , and ≤ 4 cm contrast-enhancing recurrence after initial therapy. Because of the association of ATO with QT prolongation, patients taking medication associated with QT prolongation and/or showing baseline QT abnormalities were excluded. Four dose levels (0.10, 0.15, 0.20, 0.25 mg/kg/dose) will be explored with at least 3 patients enrolled at each level. ATO was delivered intravenously over 120 minutes within 2 hours of fractionated stereotactic radiation on each of 3 consecutive days using image-guided frameless radiosurgery to total radiation doses of 22–27 Gy depending on the size of the recurrent tumor. Dose-limiting toxicity (DLT) was defined as grade 3 or 4 non-hematologic adverse events, QT prolongation >500 msec or >450 msec if associated with ventricular arrhythmias, or persistent grade 4 hematologic toxicity. Patients were monitored for hemodynamic changes in blood pressure, electrolyte abnormalities, and EKG abnormalities during the course of treatment. The patients were followed thereafter by a multidisciplinary team on a monthly basis after completion of therapy and underwent serial MRI with spectroscopy at one month and thereafter as clinically indicated.

Results: Seven patients with 10 index lesions have been enrolled to 3 of the 4 dose levels. Mean age was 58 years; median KPS was 70; 5 patients had recurrent grade IV tumors and two patients had recurrent grade III tumors. All patients completed the intended therapy. There have been no DLT; though grade 1–2 constitutional, neurologic, gastrointestinal, cardiac, and metabolic toxicities were encountered. To date, 3 patients have died due to further disease progression at 3.5, 5, and 7.5 months, respectively. The remaining 7 patients are alive at 1–12 months after treatment. Six patients with 9 lesions were available for radiographic follow-up. Among these 9 lesions treated, 3 lesions showed complete regression of the contrast enhancing abnormality; 2 lesions showed partial regression, and 4 were stable at initial radiographic follow-up.

Conclusions: ATO and fractionated stereotactic radiation appears to be well tolerated at the dose levels examined. The radiographic response is encouraging. We plan to continue further dose escalation and a phase II trial in the future.

1022 WITHDRAWN

1023 Optimizing Linac Based Stereotactic Radiotherapy of Uveal Melanomas: Seven Years Clinical Experience

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Purpose/Objective: To report on the clinical outcome of linac based stereotactic radiotherapy of uveal melanomas. Additionally, a new prototype (hardware and software) for automated eye monitoring and gated SRT using a non-invasive eye fixation technique is described.

Materials/Methods: Between June 1997 and March 2004, 158 patients suffering from uveal melanoma were treated at a LINAC with 6 MV (5 $\times$ 14Gy; 5 $\times$ 12Gy prescribed to 80% isodose) photon beams. To guarantee identical patient setup during